

CG2023 Report
Mini-Project: Single Tone Amplitude Modulation and Demodulation

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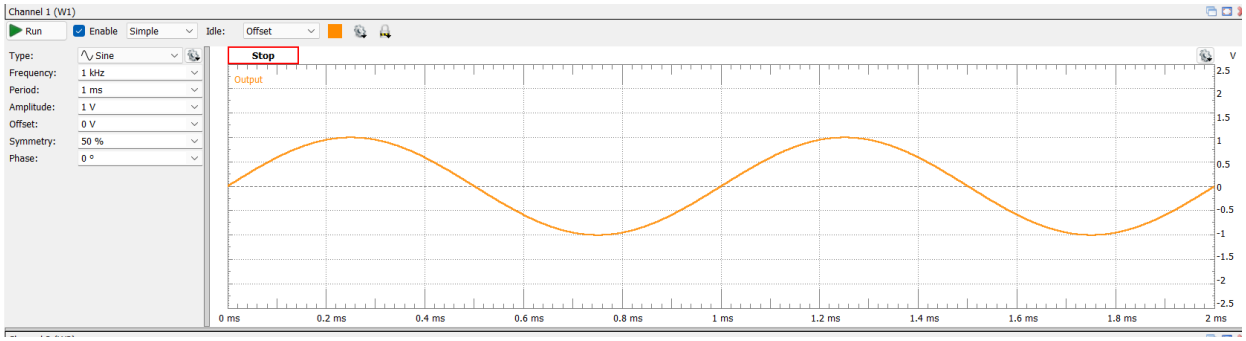
Question 1: *Plots of the waveform and spectrum of each of the message signal* $m(t)$*, carrier wave* $c(t)$*, and AM signal* $x(t)$ *using the AD 2 oscilloscope and spectrum analyzer function.*

Modulation Configuration

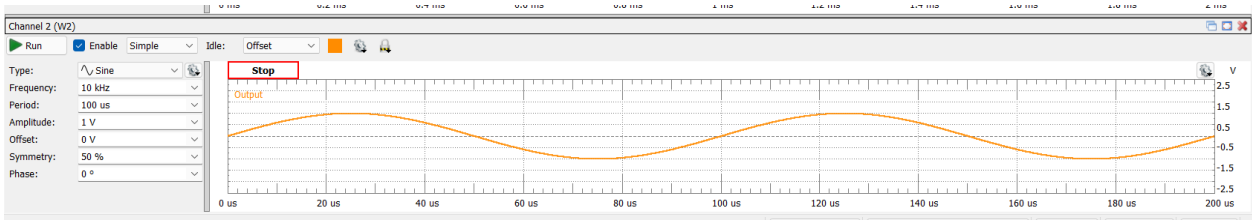
We set the frequency of f_m to be 1kHz, which is in the middle of the audible range of $f_m = [400,\ 4000]$ Hz. Since frequency of f_c needs to be at least 10 times that of f_m, we set the frequency of f_c to be exactly 10 times that of f_m, which is 10kHz. This was configured via the WaveGen feature in WaveForms.

We also ensured that the amplitude of both of the waves were 1V.

f_m waveform:

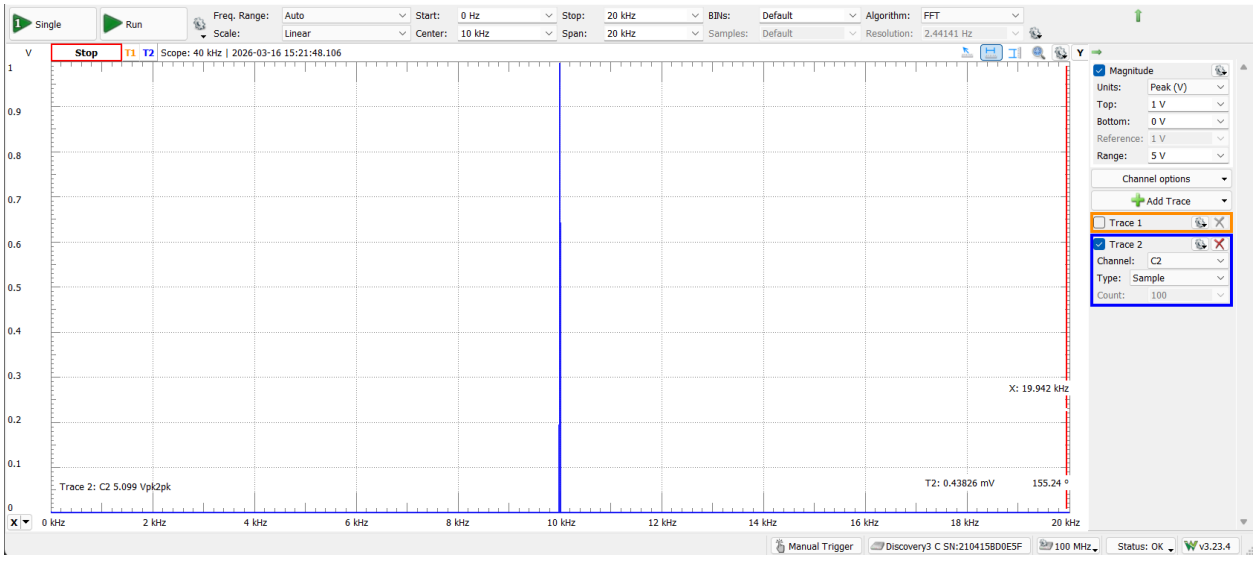


f_c waveform:



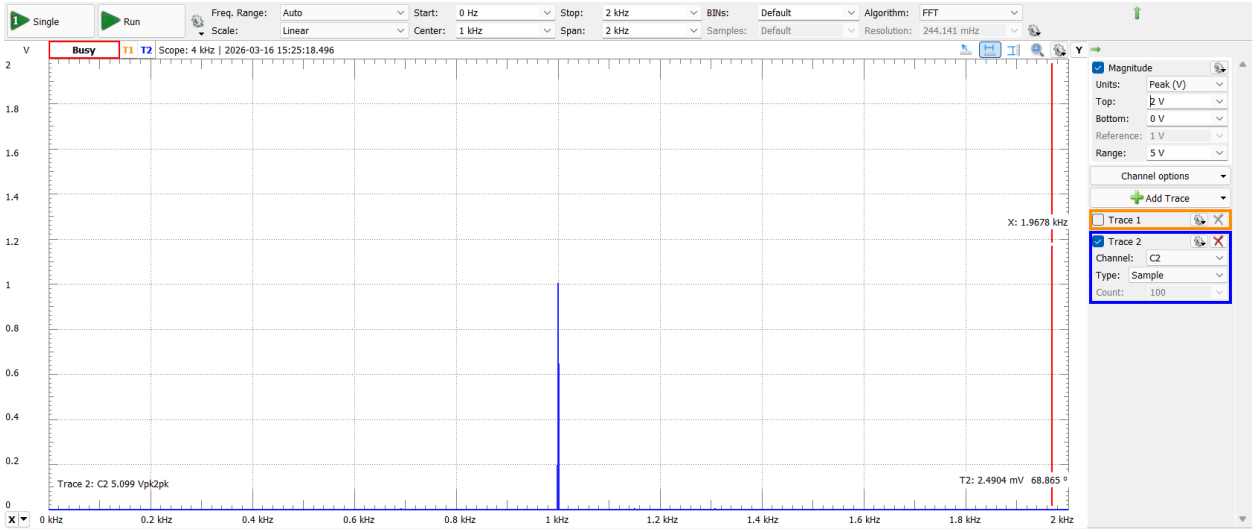
Plot of c(t) in frequency domain

There is a sharp peak that is at $f_c = 10kHz$

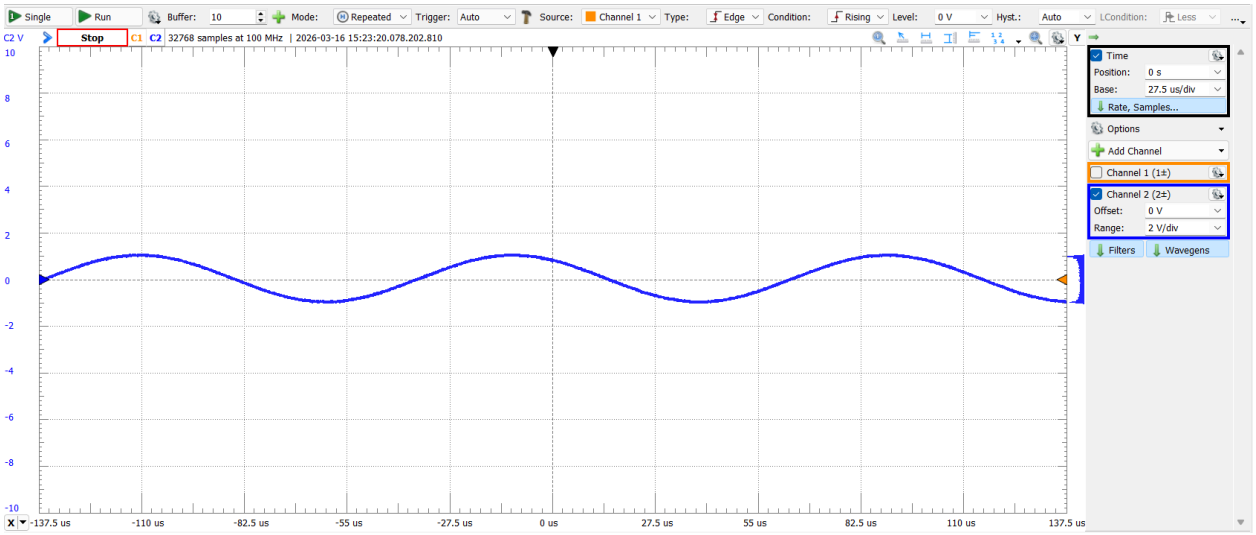


Plot of m(t) in frequency domain

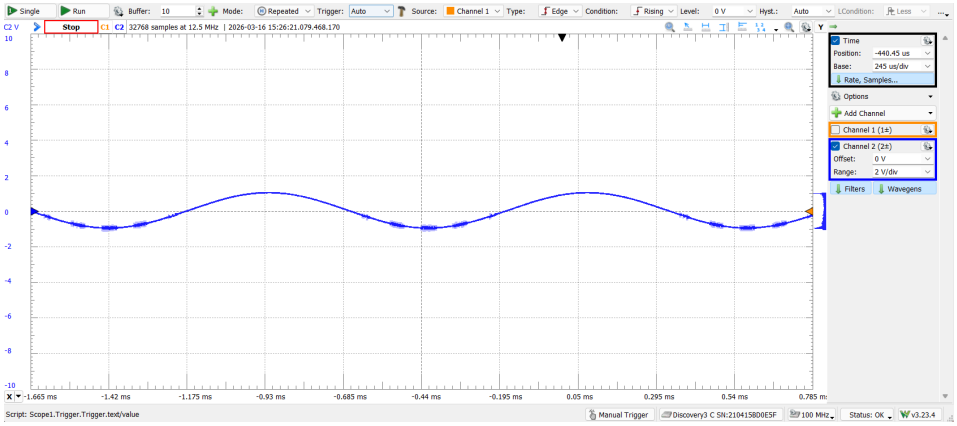
There is a sharp peak at $f_m = 1kHz$



Plot of c(t) in time domain

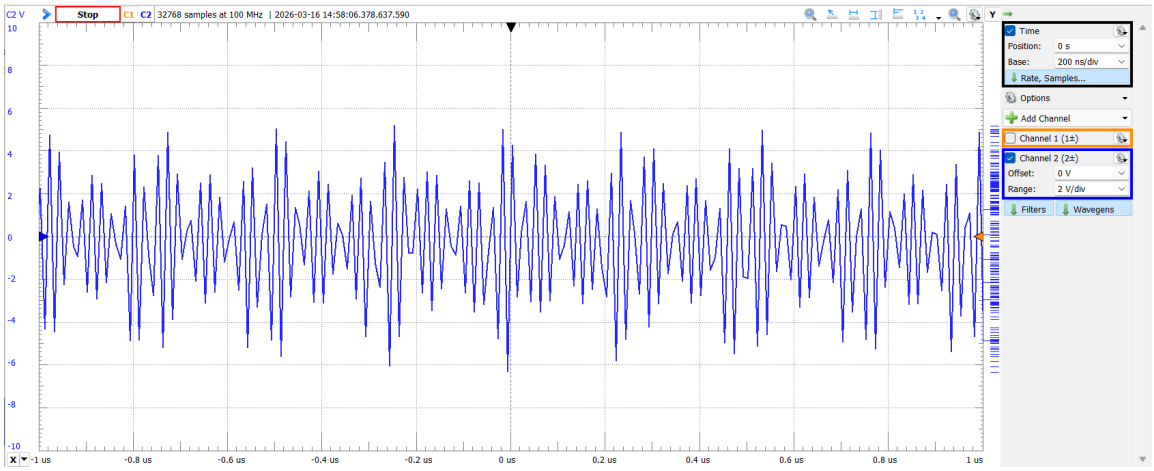


Plot of m(t) in time domain

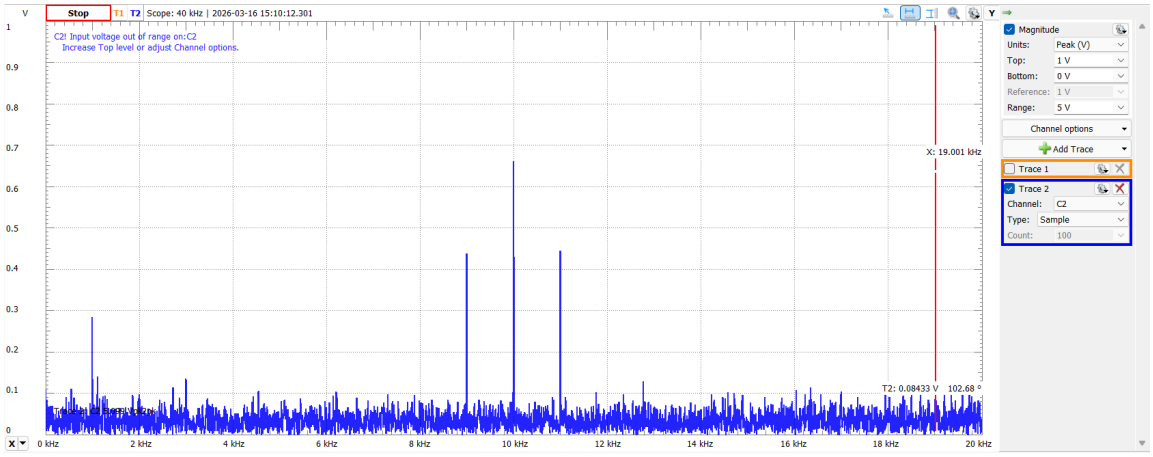


We got an AM signal of the message signal m(t), which looked like this in time and frequency domain.

Waveform of AM signal x(t) in time-domain



Waveform of AM signal x(t) in frequency domain



Question 2: *List the RC value that you have chosen for the envelope detector and explain your choice*

As you can see, there are peaks at frequency f_m, $f_c - f_m$, f_c and $f_c + f_m$, which is 1kHz, 9kHz, 10kHz, and 11kHz respectively, with the highest peak being at 10kHz.

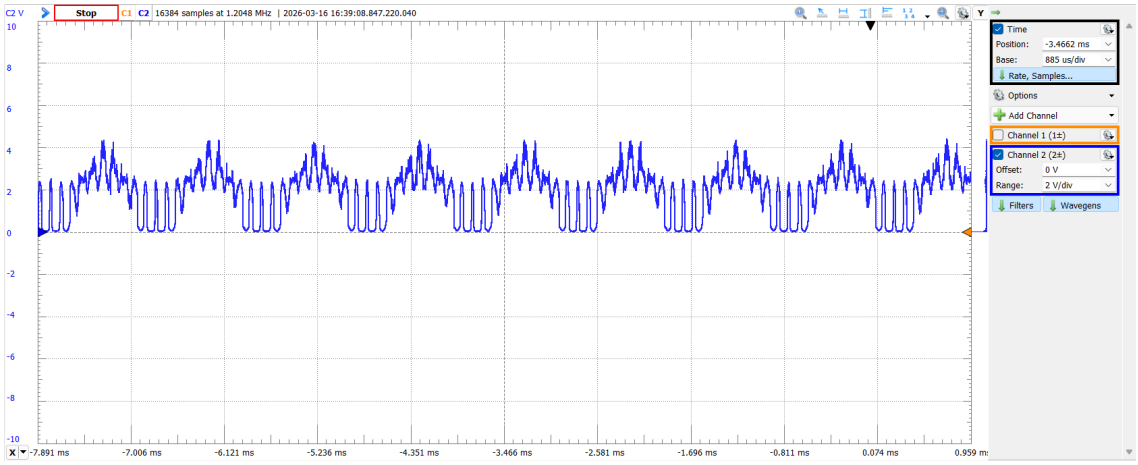
For the envelope detector, we decided to go with an RC value that fits within the criteria of $1/f_m >> \tau = RC >> 1/f_c$, which meant that it had to be much bigger than 0.0001, but much smaller than 0.001.

Through trial and error of multiple RC values, we decided on $R = 10k\Omega$ and $C = 82000pF = 82nF$, giving an RC value of 0.00082.

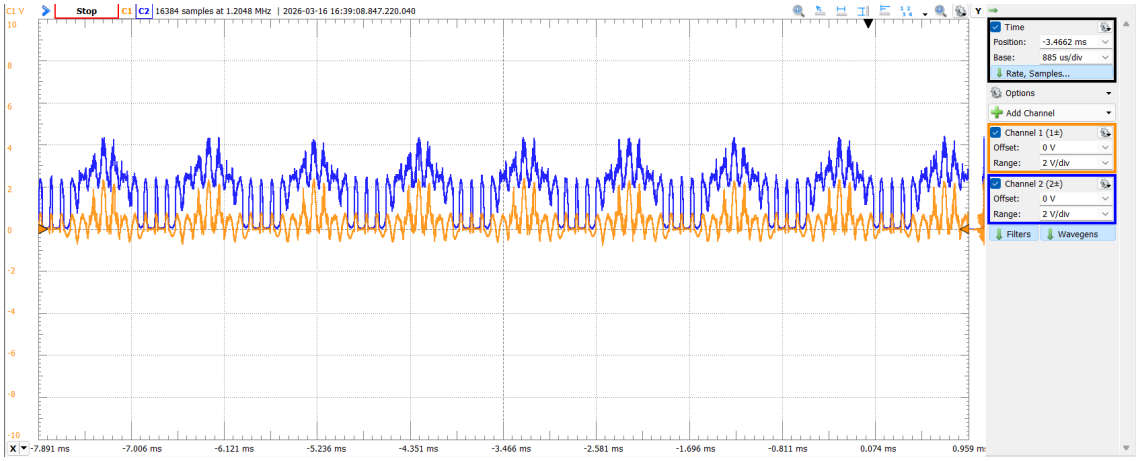
This RC value managed to give us a clear illustration of the capacitor's charging and discharging behaviour, including a well-defined peak, while maintaining a smooth waveform with minimal noise. Consequently, this set of values produced the most stable and interpretable signal for analysis.

Question 3: *Plots of the waveform and spectrum of the envelope detector output* $\widehat{m}(t)$*:*
i) without the smoothing capacitor; ii) with the smoothing capacitor, using the AD 2 oscilloscope and spectrum analyzer functions.

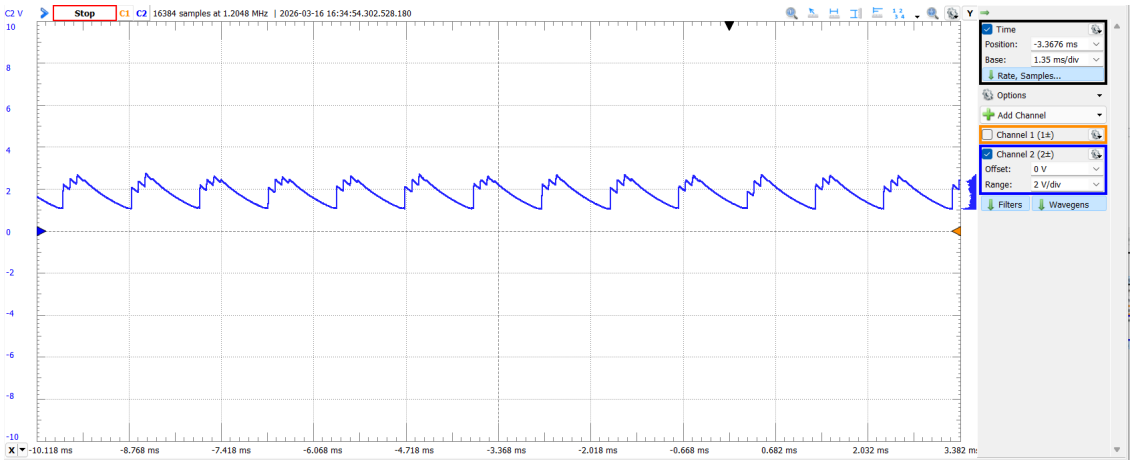
The demodulated message **without capacitor** is as such:



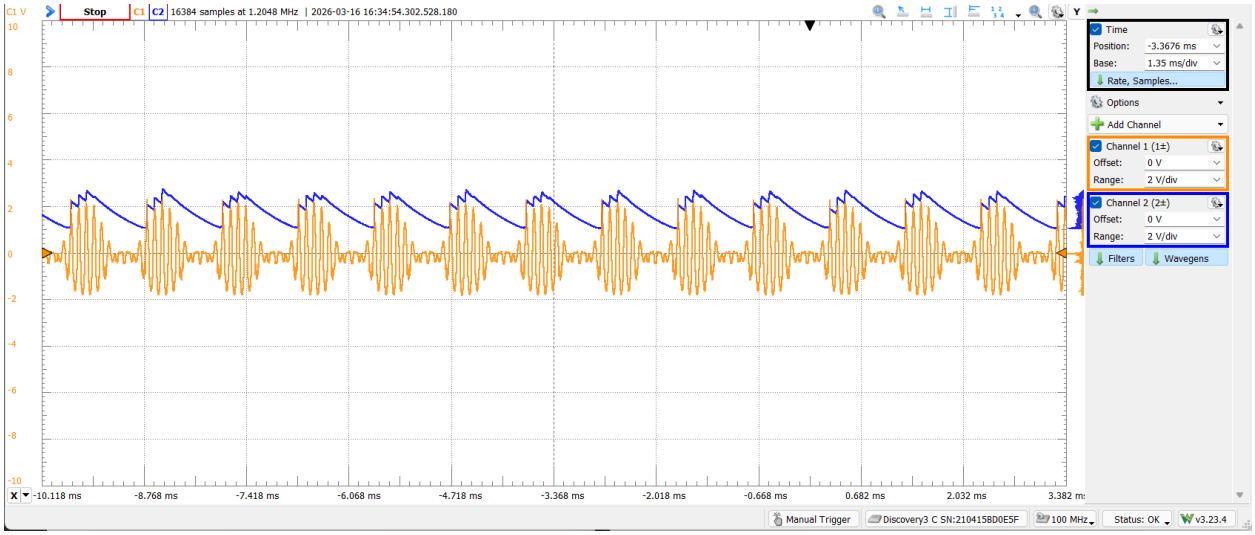
This is the demodulated message **without capacitor,** relative to the AM signal x(t):



The demodulated message **with the smoothing capacitor** is as such:



This is the demodulated message **with the smoothing capacitor**, relative to the AM signal x(t):



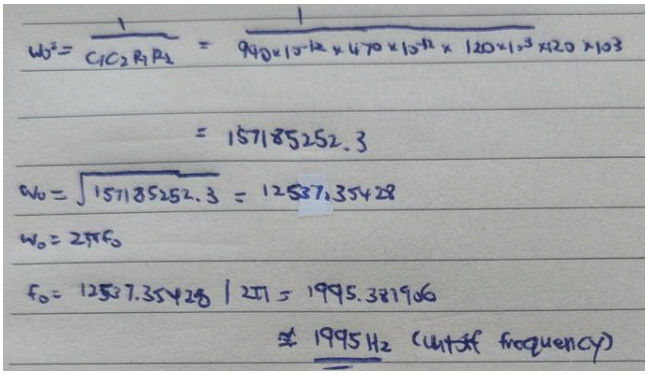
Our eventual circuit looks like this for the modulation and demodulation (paste image)

***Question 4**: The cut-off frequency and the values of R_1 R_2, C_1, and C_2 of the 2nd-order Butterworth lowpass filter.*

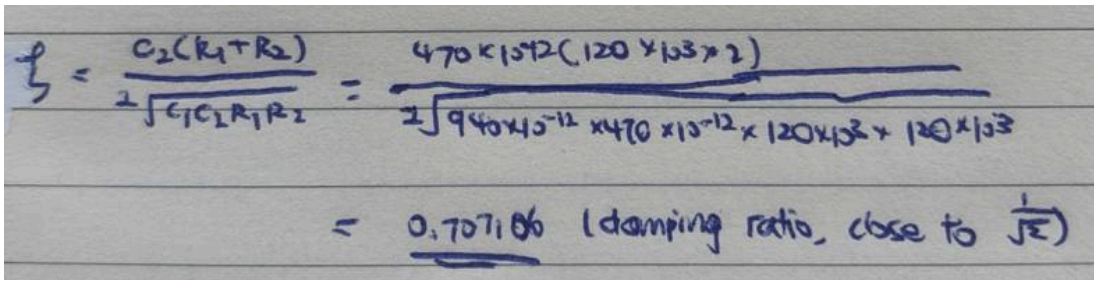
For the 2nd-order Butterworth lowpass filter, we decided on these values. We were aiming for values that were close to the ratio of $R_1 = R_2 = R$ and $C_1 = 2C_2 = 2C$, based on the resistors and capacitors found at the lab. We also had to make sure that the resistors used for the 2nd order Butterworth low-pass filter were at least 10 times larger than the resistors that were used for the modulation and demodulation of the signal, which were $10k\Omega$**.** These were the values that we decided upon for our low-pass filter:

R_1	R_2	C_1	C_2
$120k\Omega$	$120k\Omega$	$940pF$	$470pF$

Based on the calculations, the cut-off frequency should be **1995Hz** approximately, which is close to the cutoff frequency of 2000Hz.

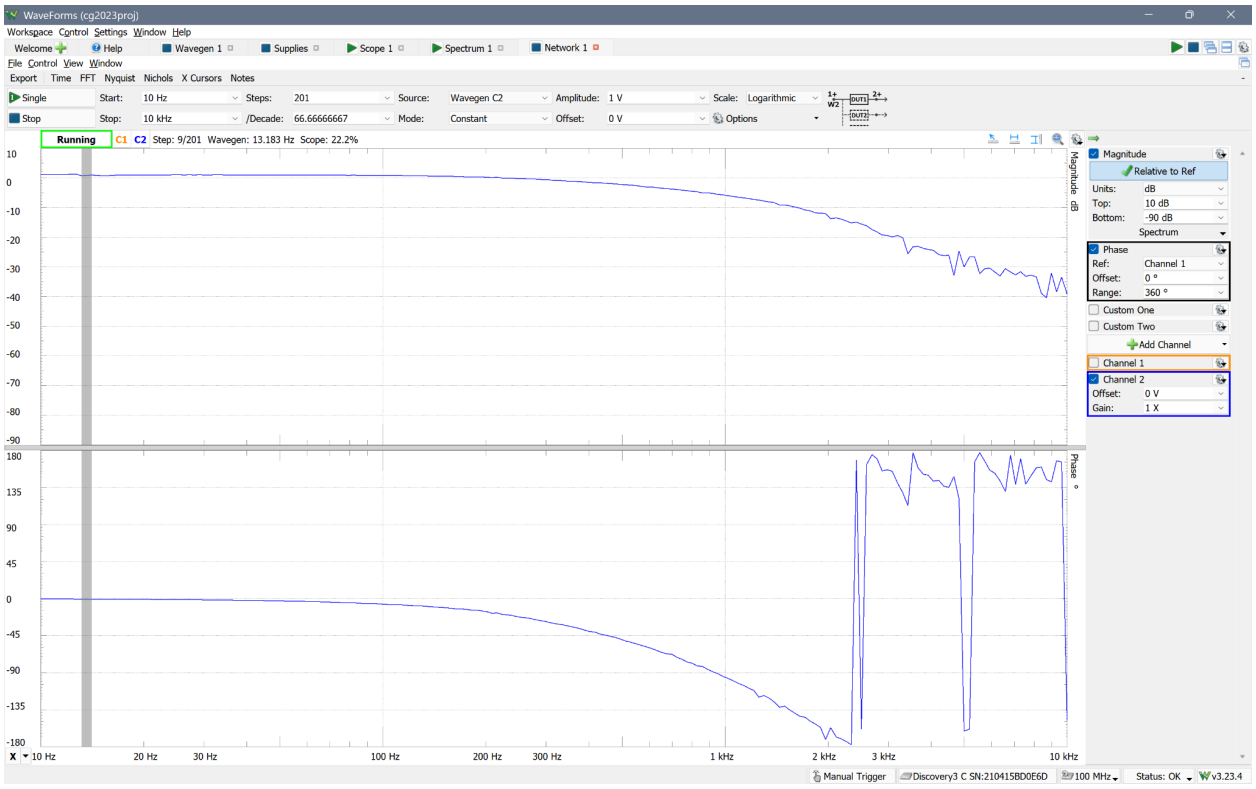


Based on our values, the damping ratio is 0.707106, which is close to the intended ratio of $1/sqrt(2)$.



***Question 5**: The Bode diagrams for your 2nd-order Butterworth lowpass.*

What we expect for the magnitude Bode diagram (top) is that it should first plateau at 0V which is the DC gain value K, and then descend linearly at a rate of 2 x -20dB/decade = **-40dB/decade only at the cutoff frequency of 1995Hz**, since it is a 2nd order Butterworth low-pass filter.
For the phase Bode diagram (bottom), it should descend linearly from 0 degrees to 180 degrees, and when it reaches the cutoff frequency of about 1995Hz, it should have a **phase shift of 180 degrees**, making the phase equal to 0 degrees after the cutoff frequency, and then plateau at 0 degrees from there.



However, the experimental behaviour was slightly different from the theoretical behaviour since it is not under ideal conditions. For the magnitude spectrum, the descent starts earlier than the cutoff frequency itself of 1995Hz. For the phase spectrum, the descent between 0Hz and the cutoff frequency of 1995Hz is not linear.

Question 6: *An explanation of how the selection of A_m impacts the difficulty of choosing the RC time constant in the envelope detector.*

The selection of A_m determines how steep the envelope's falling edge can get. This tightens or loosens the upper bound on the RC time constant.

The envelop detector should be able to handle the following to requirements:

1. RC must be small enough so capacitor discharges quickly and tracks falling edge of the envelope.
2. RC must be large enough so capacitor can hold its charge between high-frequency carrier peaks.

$$1/f_c << \tau = RC << 1/f_m$$

Mathematical Derivation:

Given that the envelope of the signal, $E(t)$, is:

$$E(t) = A_c(1 + A_m cos(2\pi f_m t))$$

$$\frac{dE(t)}{d(t)} = -A_c A_m (2\pi f_m) sin(2\pi f_m t)$$

Meanwhile the capacitor discharges through the resistor R according to the standard decay equation:

$$V_c(t) = V_0 e^{-t/RC}$$

Rate of discharge:

$$\frac{dV_c(t)}{dt} = -\frac{V_0 e^{-t/RC}}{RC} = -\frac{V_c}{RC}$$

When the high-frequency carrier wave hits a peak, the diode is on, and charges the capacitor to that peak. Here $V_c(t) = E(t)$.

Once the high-frequency carrier starts to drop, the capacitor starts discharging and the diode is turned off immediately. Since we want to ensure that the rate at which the capacitor discharges is equivalent to the rate at which the signal drops, we have to ensure the rate of discharge equals $\frac{dE(t)}{d(t)}$.

Since we want capacitor voltage to closely track the envelope, $E(t) \approx V_c(t)$:

$$\frac{V_c}{RC} \geq \left|\frac{dE(t)}{dt}\right| \rightarrow \frac{E(t)}{RC} \geq \left|\frac{dE(t)}{dt}\right|$$

$$\frac{A_c(1+A_m cos(2\pi f_m t))}{RC} \geq A_c A_m (2\pi f_m) sin(2\pi f_m t)$$

$$RC \leq \frac{1+A_m cos(2\pi f_m t)}{A_m (2\pi f_m) sin(2\pi f_m t)}$$

To guarantee the detector works at all times, we must satisfy the **absolute worst-case scenario** (the moment the envelope is falling the fastest). Using calculus, the minimum value of this function occurs when $cos(2\pi f_m t) = -A_m$.

$$RC \leq \frac{\sqrt{1-A_m^2}}{2\pi f_m A_m}$$

The message amplitude, A_m, directly determines how deep and steep the envelope's falling edge can get. As A_m approaches 1, the envelope's downward slope becomes much steeper. To accurately track this steeper drop without losing the signal, the capacitor is forced to discharge much faster.

However, the lower bound $1/f_c$ remains fixed, as RC must still be large enough to suppress high-frequency ripple from the carrier wave. Therefore, a larger A_m drastically squeezes the acceptable range of RC values between these two limits, making it much more difficult to find a time constant that perfectly balances tracking the envelope and smoothing the carrier.

Question 7: *An explanation of how the ratio f_m / f_c impacts the difficulty of choosing the RC time constant in the envelope detector.*

The envelop detector should be able to handle the following to requirements:

1. RC must be small enough so capacitor discharges quickly and tracks falling edge of the envelope.
2. RC must be large enough so capacitor can hold its charge between high-frequency carrier peaks.

$$1/f_c << \tau = RC << 1/f_m$$

Note that f_c is **significantly larger** than f_m**.** Thus, f_m / f_c is less than 1.

If the **ratio of f_m / f_c is small**, carrier is faster than envelope and will give a wide range of valid RC values. It will be easy to choose RC, envelope detector will work nice, and there's a smooth output and minimal distortion.

Example: If the ratio of f_m / f_c is small, where $f_m = 1kHz$ and $f_c = 10kHz$, there is a larger window for RC $100\mu s << RC << 1000\mu s$. And thus finding a valid RC is easier.

If the **ratio of f_m / f_c is large** and around the f_m is approximately f_c, then carrier and envelope are similar speeds, and it becomes very hard to find appropriate RC values that fit the condition.

Example: If the ratio of f_m / f_c is large, where $f_m = 5kHz$ and $f_c = 10kHz$, there is a smaller window for RC $100\mu s << RC << 200\mu s$. And thus finding a valid RC is harder.

Thus the lab manual recommended to set f_m / f_c to be lesser than 0.1, where f_c is at least 10 times f_m.

Conclusion:

In this experiment, amplitude modulation and demodulation were successfully implemented and analysed in time and frequency domains. The generated messages and carrier signals produced the expected AM waveform, with clear spectral components observed at $f_c, f_c + f_m, f_c - f_m, f_m$, confirming correct modulation behaviour.

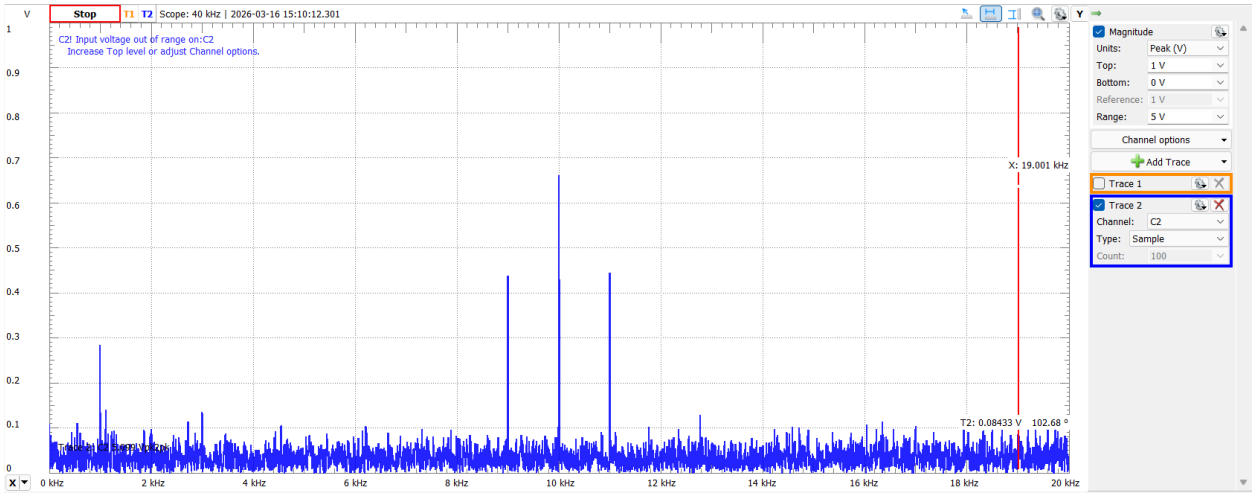
The envelope detector was then used to recover the original message signal. An appropriate RC value was found that was in between the reciprocal of f_c and f_m. This ensured that the detector effectively smoothed out the high-frequency carrier while tracking the slower variations of the envelope. The chosen values produced a stable output with minimal ripple and distortion, demonstrating the importance of carefully balancing the RC parameters.

A 2nd order Butterworth low-pass filter was designed and implemented to further refine the demodulated signal. The theoretical cutoff frequency and damping ratio closely matched the expected Butterworth characteristics. However, slight deviations were observed in the experimental Bode plots, such as earlier magnitude roll-off and non-linear phase response, which can be attributed to non-ideal components and practical limitations in the circuit.

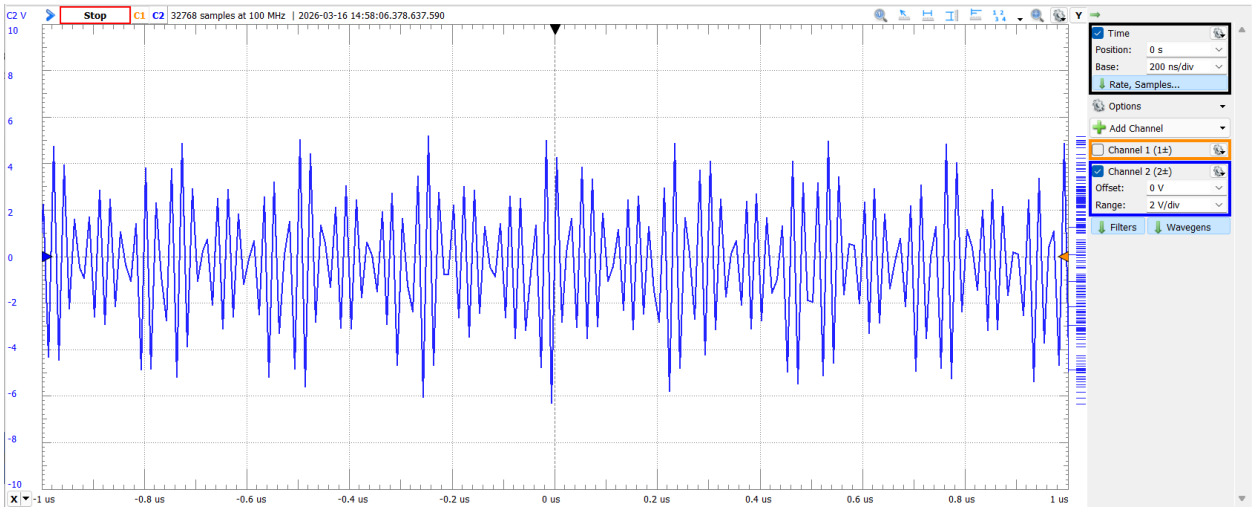
Additionally, the experiment highlighted how system parameters like modulation amplitude A_m increases the rate of change of the envelope, requiring a small RC time constant to accurately track it, while a smaller f_m/f_c ratio was ratio provides a wider range of acceptable RC values, making the design more robust.

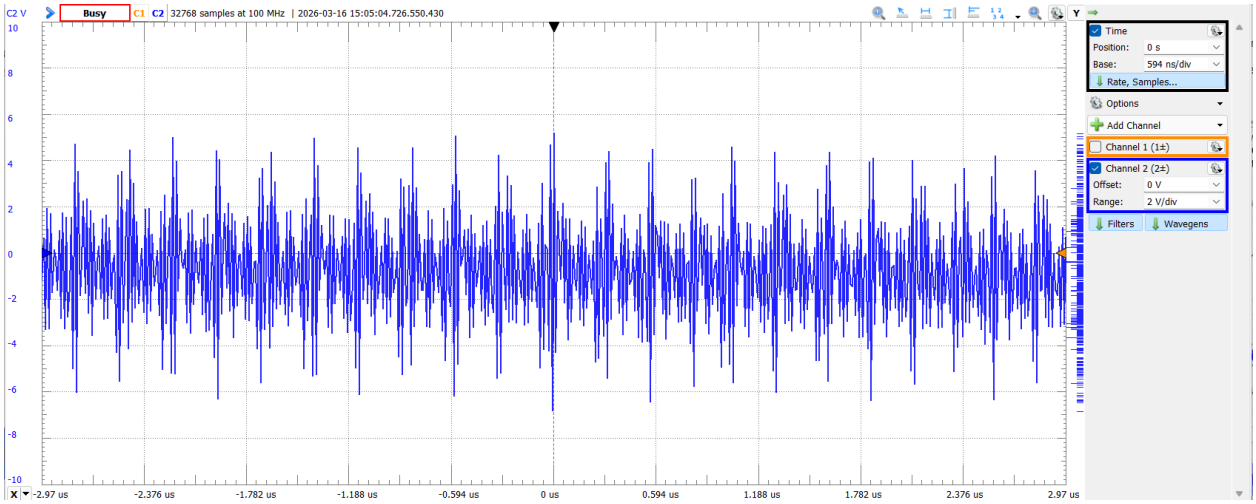
Overall, the results demonstrate the practical considerations involved in AM systems, particularly the trade-offs in selecting circuit parameters to achieve accurate signal recovery under non-ideal conditions.

Extra Graphs



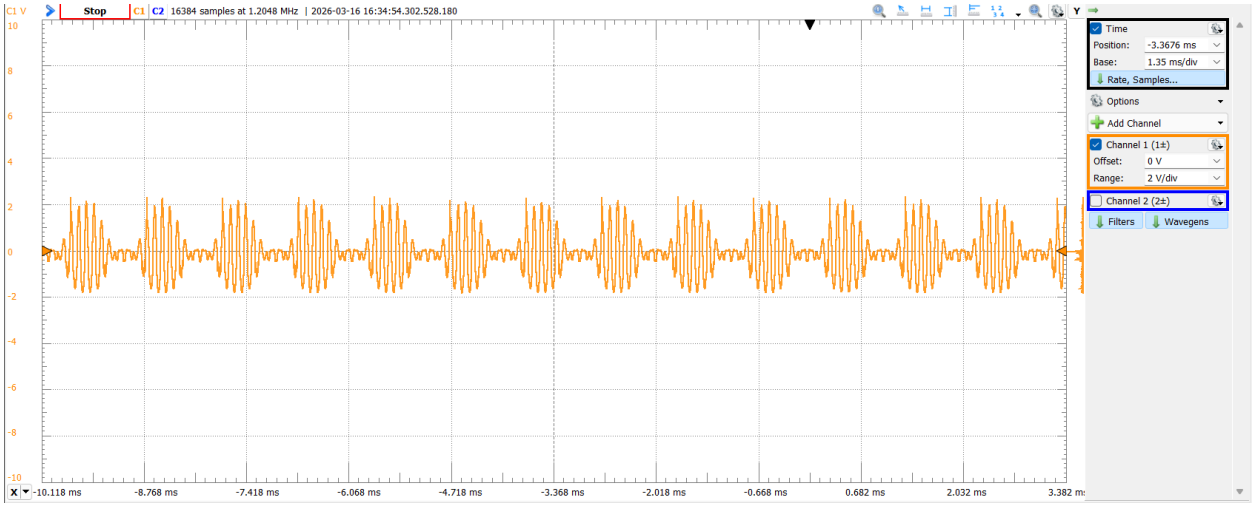
Scope



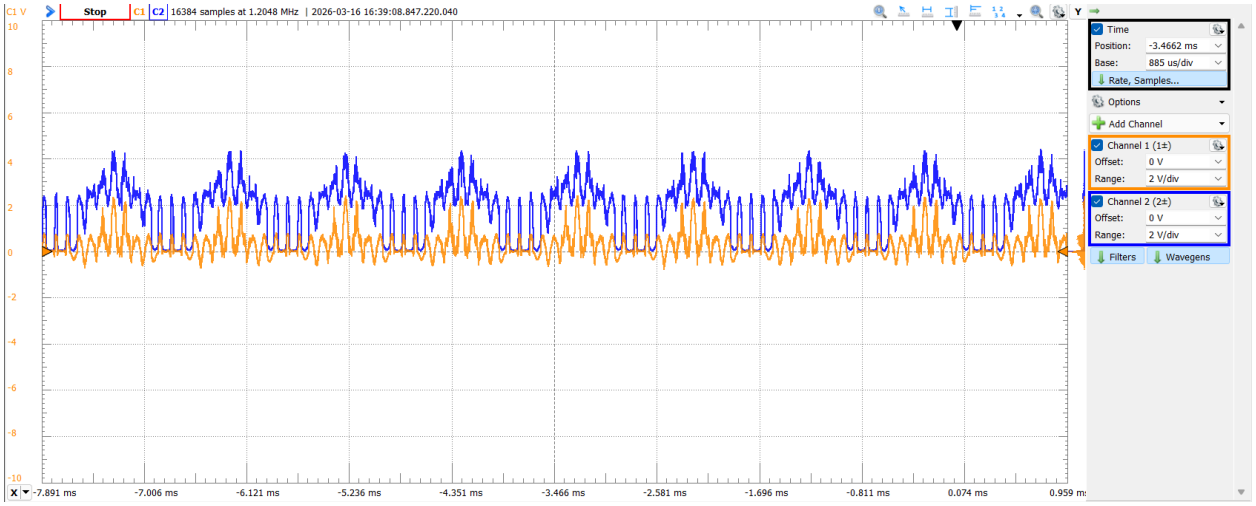


Demodulated message m(t)

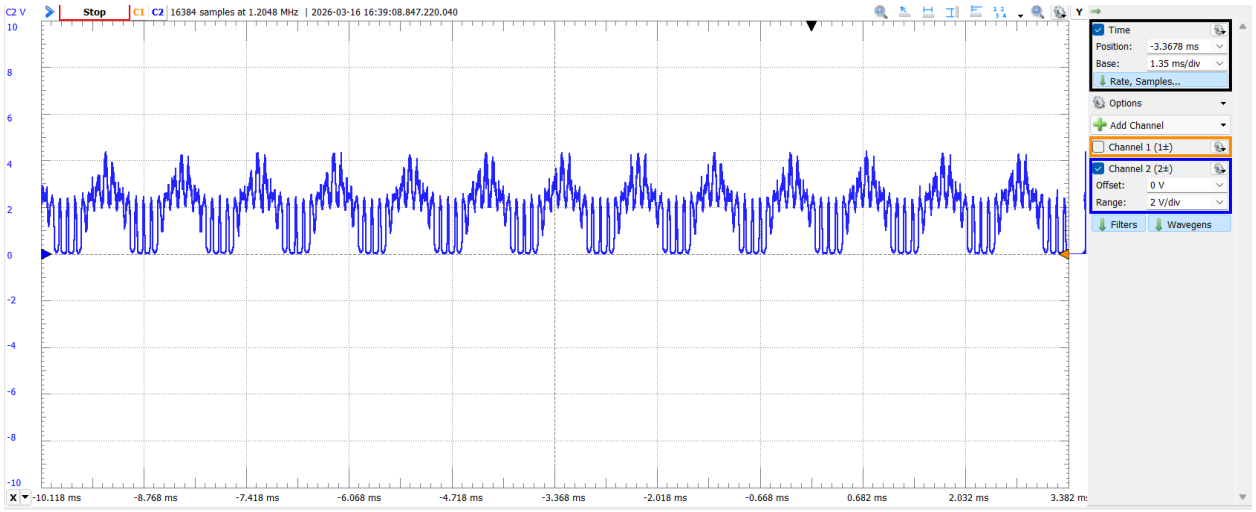
Original x(t)

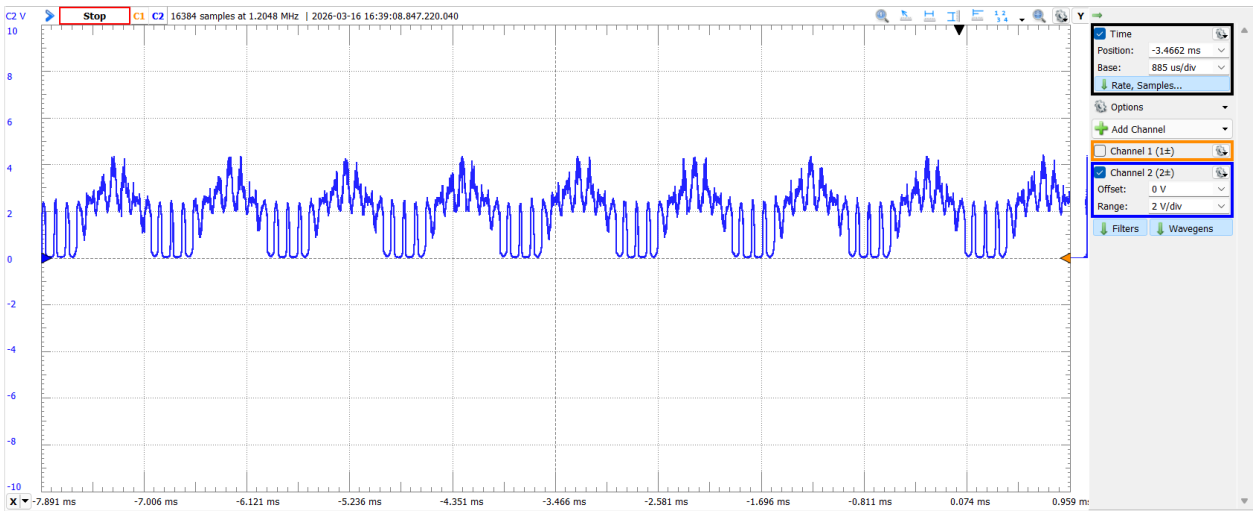


Original x(t) comparing against demodulated message without capacitor

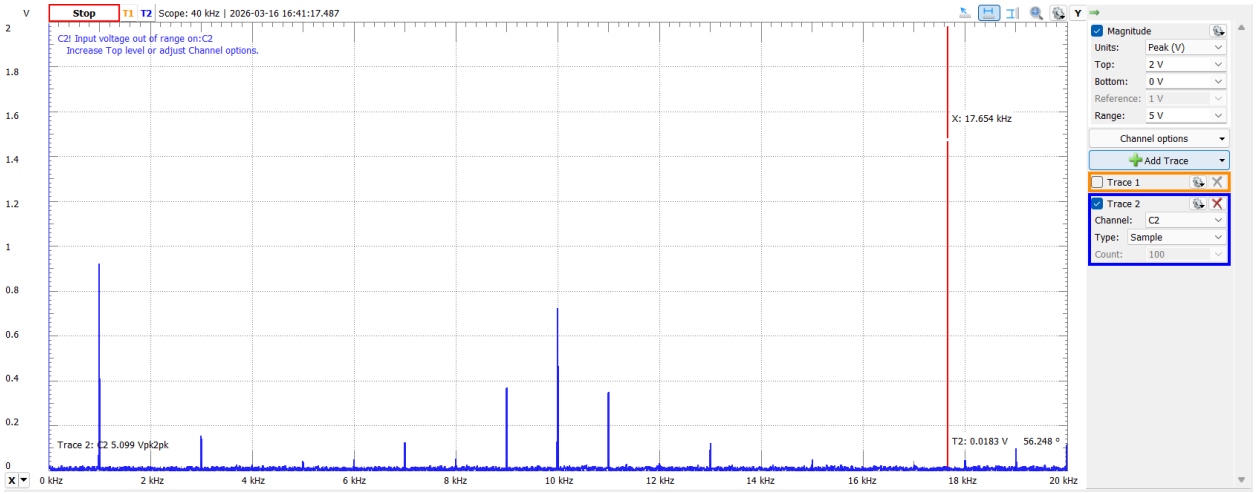


Demodulated message without capacitor

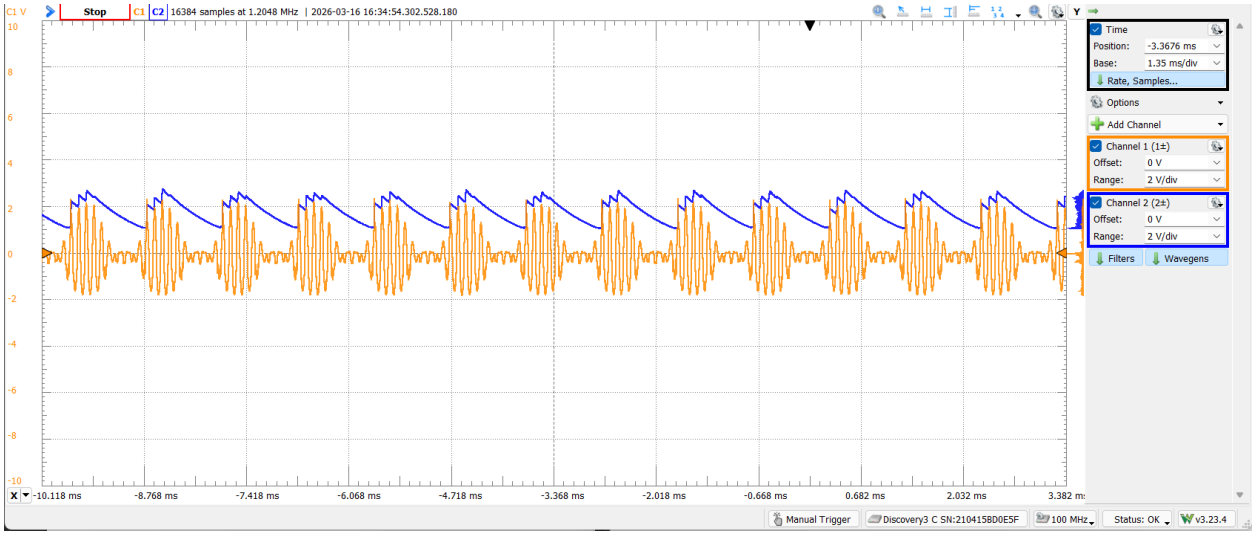




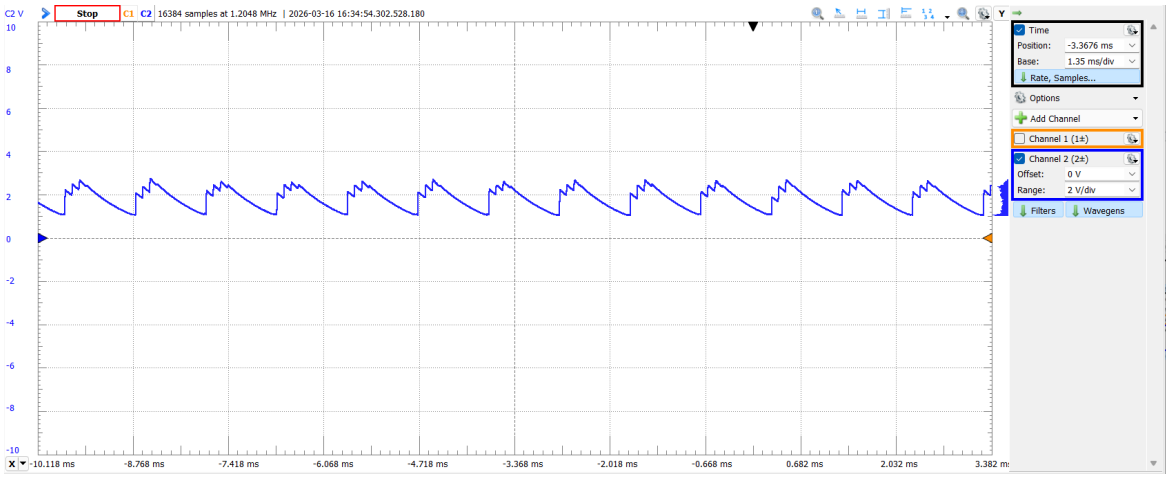
Spectrum for without capacitor



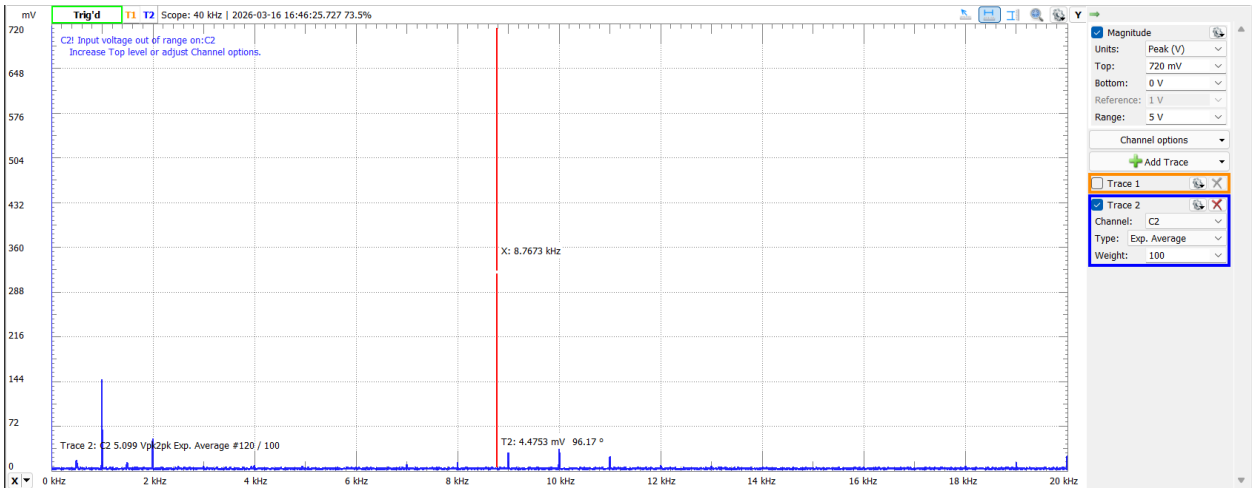
Original x(t) comparing against demodulated message with capacitor



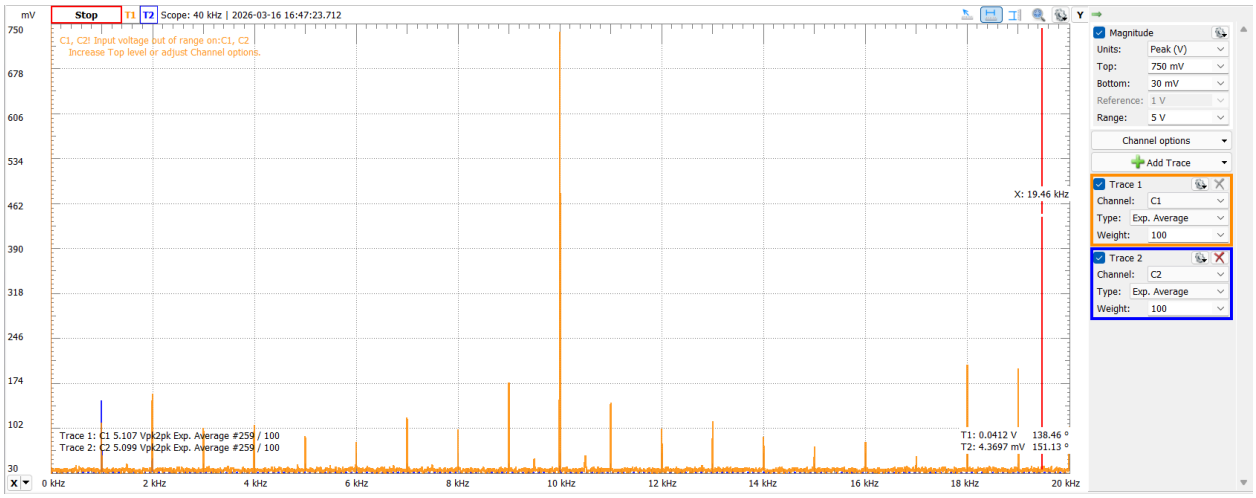
Demodulated message with capacitor



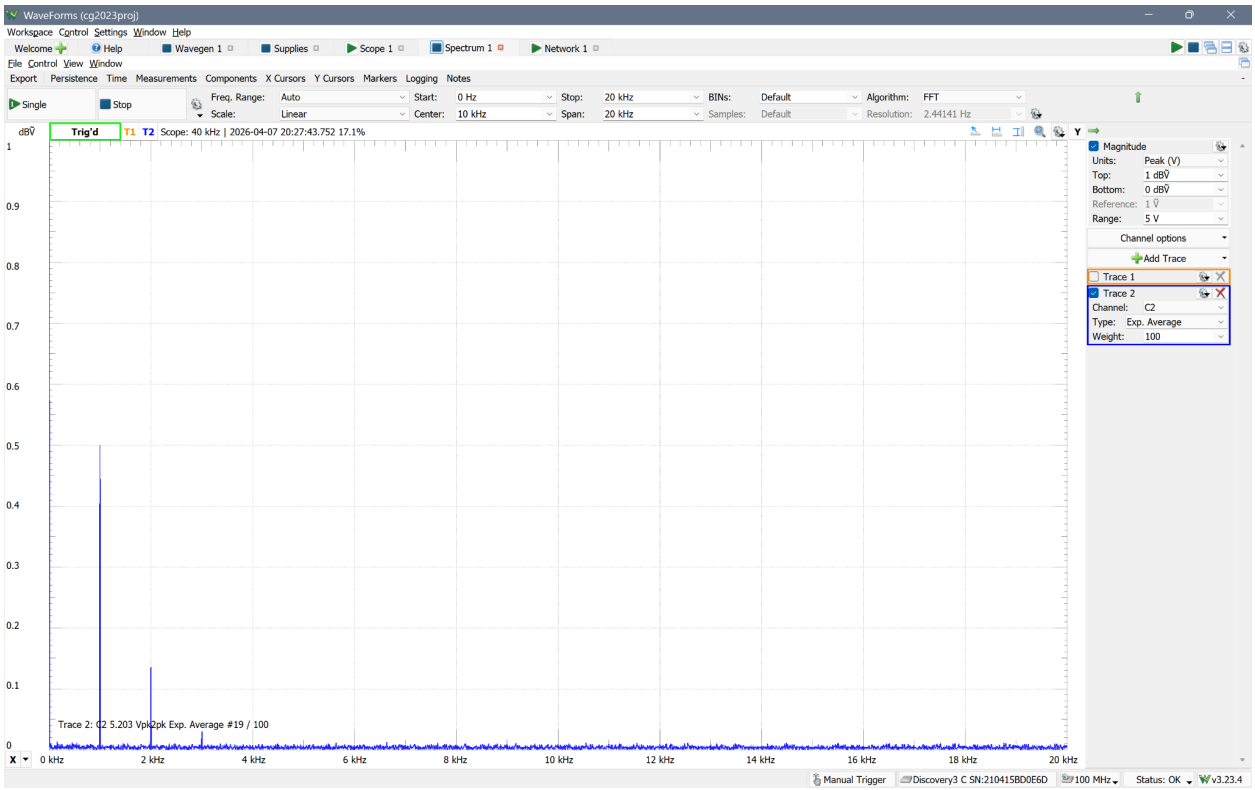
Spectrum for demodulated message (obvious peak at 1kHz, the rest is suppressed)



In comparison to W, which has the obvious peak at 10kHz. (modulation of the wave m(t) occurs)




WaveForms (cg2023proj)
Trig'd
C1
C2
16384 samples at 680.27 kHz | 2026-04-07 20:26:40.254.379.150
X2: 12.34 ms
ΔX: 23.94 ms
1/ΔX: 41.77785037 Hz
X1: -11.6 ms
Time
Position: 290.5 us
Base: 2.4 ms/div
Rate, Samples...
Options
Add Channel
Channel 1 (1±)
Offset: 0 V
Range: 1 V/div
Channel 2 (2±)
Offset: -1.3 V
Range: 1 V/div
Filters
Wavegens
Manual Trigger
Discovery3 C SN:210415BD0E6D
100 MHz
Status: OK
v3.23.4


WaveForms (cg2023proj)
Trig'd
T1
T2
Scope: 40 kHz | 2026-04-07 20:27:43.752 17.1%
Trace 2: C2 5.203 Vpk2pk Exp. Average #19 / 100
Magnitude
Units: Peak (V)
Top: 1 dBV
Bottom: 0 dBV
Range: 5 V
Channel options
Add Trace
Trace 1
Trace 2
Channel: C2
Type: Exp. Average
Weight: 100
Manual Trigger
Discovery3 C SN:210415BD0E6D
100 MHz
Status: OK
v3.23.4

Magnitude and Phase Diagram (respectively) of the 2nd order Butterworth low-pass filter

Magnitude

What actually happened: should start descending at 40dB/decade earlier than the the cutoff frequency of 2kHz

Theoretical: it should be completely straight, then only start descending at the cutoff frequency linearly at 40db/decade

Phase

What actually happened: At the cutoff frequency, it should encounter a phase shift of 180 degrees and fluctuate

Theoretical: It should descend linearly till 180 degrees, and phase shift of 180 degrees so it will “jump back up” to 0, and plateau at 0 degrees after the cutoff frequency.

